

ESAT PRACTICE TEST SERIES

ESAT Mock Test 5

Mathematics 1 · Physics · Mathematics 2

- 27 multiple-choice questions per module
- 40 minutes per module
- Non-calculator
- Hard ESAT-style problem set with full worked solutions

SOLUTION BOOK

ThrivingScholars

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Answer keys

Mathematics 1 · Physics · Mathematics 2

Mathematics 1

1	C	2	B	3	D	4	A	5	E	6	C	7	B	8	D	9	A
10	E	11	C	12	B	13	D	14	A	15	E	16	B	17	C	18	D
19	A	20	E	21	C	22	B	23	D	24	A	25	E	26	C	27	B

Physics

1	D	2	D	3	C	4	C	5	B	6	D	7	D	8	C	9	D
10	C	11	E	12	D	13	C	14	C	15	D	16	D	17	E	18	D
19	C	20	C	21	B	22	B	23	D	24	D	25	B	26	C	27	C

Mathematics 2

1	D	2	B	3	E	4	C	5	B	6	C	7	B	8	B	9	C
10	B	11	B	12	B	13	C	14	B	15	B	16	C	17	C	18	D
19	D	20	C	21	C	22	B	23	C	24	C	25	B	26	E	27	D

Question 1

Simplify fully

$$\frac{(50x^{3/2}y^{-1})^2(2x^{-1}y^{5/2})^3}{(20x^{1/2}y)^2},$$

where x and y are positive.

A $\frac{10y^3\sqrt{y}}{x}$

B $\frac{50y^3}{x}$

C $\frac{50y^3\sqrt{y}}{x}$

D $\frac{50\sqrt{y}}{x}$

E $\frac{250y^3\sqrt{y}}{x}$

WORKED SOLUTION

$$(50x^{3/2}y^{-1})^2 = 2500x^3y^{-2} \text{ and } (2x^{-1}y^{5/2})^3 = 8x^{-3}y^{15/2}.$$

$$\text{So the numerator is } 2500x^3y^{-2} \cdot 8x^{-3}y^{15/2} = 20000y^{11/2}.$$

$$\text{The denominator is } (20x^{1/2}y)^2 = 400xy^2.$$

Therefore

$$\frac{20000y^{11/2}}{400xy^2} = \frac{50y^{7/2}}{x} = \boxed{\frac{50y^3\sqrt{y}}{x}}.$$

Question 2

The price of item P is reduced by 20%, and then a sales tax of 5% is applied to the reduced price.

The price of item Q is increased by 10%, and then reduced by 10%.

After these two-step changes, the final prices of P and Q are equal.

What was the ratio of the original price of P to the original price of Q ?

A 28 : 33

B 33 : 28

C 11 : 10

D 6 : 5

E 99 : 80

WORKED SOLUTION

Final price of P is $0.8 \times 1.05P = 0.84P$.

Final price of Q is $1.1 \times 0.9Q = 0.99Q$.

So $0.84P = 0.99Q$, giving

$$\frac{P}{Q} = \frac{0.99}{0.84} = \frac{99}{84} = \frac{33}{28}.$$

Question 3

For real x , define

$$E = \frac{\frac{1}{x-3} - \frac{1}{x+2}}{\frac{1}{x-3} + \frac{1}{x+2}}.$$

Given that $x > 3$ and

$$E + \frac{1}{E} = \frac{37}{6},$$

what is the value of x ?

A $\frac{11}{2}$

B 8

C 12

D $\frac{31}{2}$

E 18

WORKED SOLUTION

Simplify the fraction:

$$E = \frac{(x+2) - (x-3)}{(x+2) + (x-3)} = \frac{5}{2x-1}.$$

Put $t = E$. Then $t + 1/t = 37/6$, giving

$$6t^2 - 37t + 6 = (6t-1)(t-6) = 0.$$

Thus $t = 1/6$ or 6 . Since $x > 3$, $0 < E < 1$, so $E = 1/6$.

$$\frac{5}{2x-1} = \frac{1}{6} \Rightarrow x = \frac{31}{2}.$$

Question 4

How many integer values of x satisfy all three conditions?

- $-5 \leq x \leq 5$
- $\frac{x^2 + 3x - 10}{x^2 - 1}$ is defined
- $\frac{x^2 + 3x - 10}{x^2 - 1} \leq 1$

A 6

B 5

C 4

D 7

E 8

WORKED SOLUTION

Bring everything to one side:

$$\frac{x^2 + 3x - 10}{x^2 - 1} - 1 \leq 0 \Rightarrow \frac{3x - 9}{(x - 1)(x + 1)} \leq 0.$$

So we solve

$$\frac{x - 3}{(x - 1)(x + 1)} \leq 0, \quad x \neq \pm 1.$$

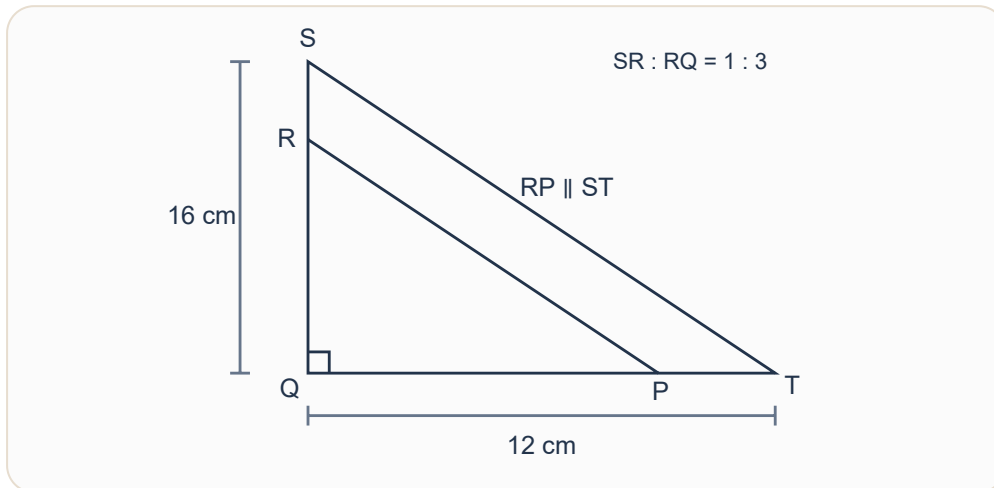
A sign chart gives $(-\infty, -1) \cup (1, 3]$.

Intersecting with $-5 \leq x \leq 5$ gives $-5, -4, -3, -2, 2, 3$, so there are **6** values.

Question 5

Triangle SQT is right-angled at Q , with $SQ = 16$ cm and $QT = 12$ cm.

Point R lies on SQ such that $SR : RQ = 1 : 3$. Through R , a line parallel to ST meets QT at P .



What is the area of quadrilateral $SRPT$?

A 24 cm²B 30 cm²C 36 cm²D 40 cm²E 42 cm²**WORKED SOLUTION**

The area of the whole triangle is $\frac{1}{2} \cdot 16 \cdot 12 = 96$.

Since $SR : RQ = 1 : 3$, we get $RQ = \frac{3}{4} \cdot 16 = 12$.

Because $RP \parallel ST$, triangles QRP and QST are similar with scale factor $\frac{QR}{QS} = \frac{12}{16} = \frac{3}{4}$.

So $QP = \frac{3}{4} \cdot 12 = 9$. Then area of triangle QRP is $\frac{1}{2} \cdot 12 \cdot 9 = 54$.

Hence quadrilateral $SRPT$ has area $96 - 54 = \boxed{42 \text{ cm}^2}$.

Question 6

The n th term of sequence T is $4n - 3$.

The n th term of sequence V is $7n + p$, where p is an integer and $0 \leq p < 7$.

The sequences have consecutive common terms **53** and **81**.

What is the value of p ?

A 1

B 2

C 4

D 5

E 6

WORKED SOLUTION

Using the common term **53**,

$$53 = 7n + p.$$

Since $53 \equiv 4 \pmod{7}$, and $0 \leq p < 7$, we must have $\boxed{p = 4}$.

Question 7

Two solid cylinders P and Q are defined using positive numbers x and y , where $x > y$.

- Cylinder P has diameter x and height y .
- Cylinder Q has diameter y and height x .

The total surface area of P is $\frac{8}{5}$ of the total surface area of Q .

The volume of P exceeds the volume of Q by $32\pi \text{ cm}^3$.

What is the value of $x + y$?

A 10

B 12

C 14

D 16

E 18

WORKED SOLUTION

For a cylinder of diameter d and height h , total surface area is $2\pi\left(\frac{d}{2}\right)^2 + 2\pi\left(\frac{d}{2}\right)h$.

Thus

$$A_P = \pi \left(\frac{x^2}{2} + xy \right), \quad A_Q = \pi \left(\frac{y^2}{2} + xy \right).$$

Set $x = ky$. Then

$$\frac{k^2 + 2k}{1 + 2k} = \frac{8}{5}.$$

So $5k^2 - 6k - 8 = 0$, giving the positive root $k = 2$. Hence $x = 2y$.

Now

$$V_P - V_Q = \frac{\pi}{4}(x^2y - y^2x) = \frac{\pi}{4}xy(x - y) = 32\pi.$$

Substitute $x = 2y$:

$$\frac{\pi}{4}(2y)(y)(y) = 32\pi \Rightarrow \frac{y^3}{2} = 32 \Rightarrow y^3 = 64 \Rightarrow y = 4.$$

Hence $x = 8$, so $x + y = \boxed{12}$.

Question 8

Pat and Alex have a combined total of £84. The ratio of Pat's money to Alex's money is 5 : 2.

They each spend the same amount on sweets.

After this, Pat receives £6 and Alex receives £1. The ratio of their new amounts is then 3 : 1.

How much did each of them spend?

A £3

B £4

C £4.25

D £4.50

E £6

WORKED SOLUTION

Initially the amounts are 60 and 24.

If each spends s , then afterwards they have $60 - s$ and $24 - s$. After receiving the extra money, they have $66 - s$ and $25 - s$.

So

$$66 - s = 3(25 - s) \Rightarrow 66 - s = 75 - 3s \Rightarrow 2s = 9 \Rightarrow s = 4.5.$$

Each spent £4.50.

Question 9

The circle $x^2 + y^2 = 25$ has centre at the origin.

A tangent to the circle passes through the point $(1, 7)$ and has positive gradient.

What is the equation of the tangent?

A $3x - 4y + 25 = 0$

B $4x - 3y + 25 = 0$

C $3x + 4y - 25 = 0$

D $4x + 3y - 25 = 0$

E $y = \frac{4}{3}x + \frac{17}{3}$

WORKED SOLUTION

Let the tangent be $y = mx + c$. Since it passes through $(1, 7)$, we have $c = 7 - m$.

The distance from the origin to the tangent must equal the radius 5:

$$\frac{|c|}{\sqrt{m^2 + 1}} = 5.$$

So

$$(7 - m)^2 = 25(m^2 + 1) \Rightarrow 12m^2 + 7m - 12 = 0 = (4m - 3)(3m + 4).$$

The positive gradient is $m = \frac{3}{4}$. Then $c = 7 - \frac{3}{4} = \frac{25}{4}$.

Hence

$$y = \frac{3}{4}x + \frac{25}{4} \iff \boxed{3x - 4y + 25 = 0}.$$

Question 10

Two fair dice X and Y are available.

- The faces on X are labelled 1, 1, 2, 3, 4, 5.
- The faces on Y are labelled 2, 3, 3, 4, 5, 6.

One of the dice is chosen at random and rolled twice.

Given that the total shown is 7, what is the probability that the chosen die was X ?

A $\frac{1}{3}$

B $\frac{3}{8}$

C $\frac{1}{2}$

D $\frac{3}{5}$

E $\frac{2}{5}$

WORKED SOLUTION

For die X , the ordered outcomes summing to 7 are (2, 5), (5, 2), (3, 4), (4, 3): 4 outcomes out of 36.

So $P(7 | X) = \frac{1}{9}$.

For die Y , 3 appears twice, so the ordered outcomes summing to 7 are 6 in total. Hence

$P(7 | Y) = \frac{1}{6}$.

With equal prior probability of choosing each die,

$$P(X | 7) = \frac{\frac{1}{2} \cdot \frac{1}{9}}{\frac{1}{2} \cdot \frac{1}{9} + \frac{1}{2} \cdot \frac{1}{6}} = \frac{\frac{1}{9}}{\frac{1}{9} + \frac{1}{6}} = \frac{2}{5}.$$

Question 11

The equation $2^x + 2^{2-x} = 5$ has two real solutions.

What is the larger possible value of x ?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Let $t = 2^x$. Then $2^{2-x} = \frac{4}{t}$, so

$$t + \frac{4}{t} = 5 \Rightarrow t^2 - 5t + 4 = 0 = (t - 1)(t - 4).$$

Thus $t = 1$ or $t = 4$, so $x = 0$ or $x = 2$. The larger value is **2**.

Question 12

Suppose x and y satisfy

$$x = y + \frac{4}{y},$$

where $y > 0$ and $x > 4$.

Which of the following is the larger possible value of y in terms of x ?

A $\frac{x - \sqrt{x^2 - 16}}{2}$

B $\frac{x + \sqrt{x^2 - 16}}{2}$

C $\frac{x + \sqrt{x^2 + 16}}{2}$

D $\frac{2}{x + \sqrt{x^2 - 16}}$

E $\frac{4}{x + \sqrt{x^2 - 16}}$

WORKED SOLUTION

Multiply by y :

$$xy = y^2 + 4 \Rightarrow y^2 - xy + 4 = 0.$$

Using the quadratic formula,

$$y = \frac{x \pm \sqrt{x^2 - 16}}{2}.$$

The larger possible value uses the positive sign, so

$$y = \frac{x + \sqrt{x^2 - 16}}{2}.$$

Question 13

A solid cylinder has radius r cm and height h cm.

A cube has side length $\sqrt{\pi} r$ cm.

The total surface area of the cylinder is equal to four times the total surface area of the cube.

What is $\frac{h}{r}$?

A 7

B 9

C 10

D 11

E 12

WORKED SOLUTION

The cylinder's total surface area is $2\pi r^2 + 2\pi r h = 2\pi r(r + h)$.

The cube's total surface area is $6(\sqrt{\pi} r)^2 = 6\pi r^2$.

Given the cylinder equals four times the cube,

$$2\pi r(r + h) = 24\pi r^2.$$

Cancel $2\pi r$:

$$r + h = 12r \Rightarrow h = 11r.$$

Therefore $\frac{h}{r} = \boxed{11}$.

Question 14

Consider the list

$$1, 2, x, x + 1, 2x - 1, 15,$$

where $2 < x < 8$.

The mean of the six numbers is one more than the median.

What is the value of x ?

A $\frac{9}{2}$

B 4

C 5

D $\frac{11}{2}$

E 6

WORKED SOLUTION

Since $x > 2$, the list is already ordered. So the median is

$$\frac{x + (x + 1)}{2} = x + \frac{1}{2}.$$

The mean is one more than this, so the mean is $x + \frac{3}{2}$.

The mean is also

$$\frac{1 + 2 + x + (x + 1) + (2x - 1) + 15}{6} = \frac{4x + 18}{6}.$$

Therefore

$$\frac{4x + 18}{6} = x + \frac{3}{2} \Rightarrow 4x + 18 = 6x + 9 \Rightarrow 2x = 9 \Rightarrow \boxed{x = \frac{9}{2}}.$$

Question 15

A cyclist rides from A to B uphill at 12 km h^{-1} , then returns from B to A along the same route at 18 km h^{-1} .

She spends exactly 6 minutes stationary at B .

The total time for the complete trip is 46 minutes.

What is the distance from A to B ?

A 3.6 km

B 4.0 km

C 4.2 km

D 4.5 km

E 4.8 km

WORKED SOLUTION

Excluding the 6-minute stop, the travel time is 40 minutes = $\frac{2}{3}$ hours.

If the one-way distance is d km, then

$$\frac{d}{12} + \frac{d}{18} = \frac{2}{3}.$$

Since $\frac{1}{12} + \frac{1}{18} = \frac{5}{36}$,

$$d \cdot \frac{5}{36} = \frac{2}{3} \Rightarrow d = \frac{2}{3} \cdot \frac{36}{5} = \frac{24}{5} = 4.8.$$

So the distance is **4.8 km**.

Question 16

Given that

$$y = \frac{2 \sin 15^\circ}{\cos 30^\circ},$$

what is the value of y^2 ?

A $\frac{4 - 2\sqrt{3}}{3}$

B $\frac{8 - 4\sqrt{3}}{3}$

C $\frac{8 + 4\sqrt{3}}{3}$

D $2 - \sqrt{3}$

E $4 - 2\sqrt{3}$

WORKED SOLUTION

Use

$$\sin 15^\circ = \sin(45^\circ - 30^\circ) = \frac{\sqrt{6} - \sqrt{2}}{4}, \quad \cos 30^\circ = \frac{\sqrt{3}}{2}.$$

Hence

$$y = \frac{2 \cdot \frac{\sqrt{6} - \sqrt{2}}{4}}{\frac{\sqrt{3}}{2}} = \frac{\sqrt{6} - \sqrt{2}}{\sqrt{3}}.$$

Therefore

$$y^2 = \frac{(\sqrt{6} - \sqrt{2})^2}{3} = \frac{6 + 2 - 2\sqrt{12}}{3} = \frac{8 - 4\sqrt{3}}{3}.$$

Question 17

The variable t is inversely proportional to the square of w , where $w > 0$.

When $w = 2 \times 10^3$, the value of t is 4.5×10^{-5} .

What is the value of w when $t = 8 \times 10^{-6}$?

A 7.5×10^3

B $3\sqrt{10} \times 10^2$

C $1.5\sqrt{10} \times 10^3$

D $2\sqrt{10} \times 10^3$

E 4.5×10^3

WORKED SOLUTION

Since $t = \frac{k}{w^2}$, use the first condition:

$$k = (4.5 \times 10^{-5})(2 \times 10^3)^2 = (4.5 \times 10^{-5})(4 \times 10^6) = 180.$$

Now solve

$$8 \times 10^{-6} = \frac{180}{w^2} \Rightarrow w^2 = 2.25 \times 10^7.$$

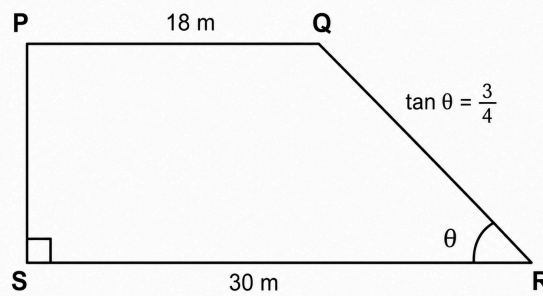
So

$$w = \sqrt{2.25 \times 10^7} = 1.5\sqrt{10} \times 10^3.$$

Question 18

$PQRS$ is a trapezium in which $PQ \parallel SR$ and $PS \perp SR$.

Also, $PQ = 18 \text{ m}$, $SR = 30 \text{ m}$, and $\tan \angle QRS = \frac{3}{4}$.



What is the area of the trapezium?

A 180 m^2

B 192 m^2

C 204 m^2

D 216 m^2

E 240 m^2

WORKED SOLUTION

The difference between the parallel sides is $30 - 18 = 12$, which is the horizontal run of the sloping side.

If the height is h , then

$$\tan \angle QRS = \frac{h}{12} = \frac{3}{4} \Rightarrow h = 9.$$

So the area is

$$\frac{1}{2}(18 + 30)(9) = 24 \cdot 9 = \boxed{216 \text{ m}^2}.$$

Question 19

The equation $x^2 + bx + 8 = 0$ has two real roots whose positive difference is 2. The sum of the roots is negative.

What is the value of b ?

A 6

B 5

C 4

D -6

E -4

WORKED SOLUTION

If the roots are r and s , then $(r - s)^2 = (r + s)^2 - 4rs$.

Here $r + s = -b$ and $rs = 8$. Since the difference is 2,

$$4 = b^2 - 32 \Rightarrow b^2 = 36.$$

So $b = \pm 6$. The sum of the roots is negative, so $-b < 0 \Rightarrow b > 0$. Hence $b = 6$.

Question 20

A rectangle has length $(2x - 1)$ cm and width $(x + 1)$ cm.

A larger rectangle is made by adding 3 cm to both the length and the width.

The larger rectangle has twice the area of the original rectangle.

What is the perimeter of the original rectangle?

A 24 cm

B 26 cm

C 28 cm

D 29 cm

E 30 cm

WORKED SOLUTION

Original area is $(2x - 1)(x + 1)$. New dimensions are $2x + 2$ and $x + 4$, so the new area is $(2x + 2)(x + 4)$.

Given new area is twice old area,

$$(2x + 2)(x + 4) = 2(2x - 1)(x + 1).$$

This simplifies to

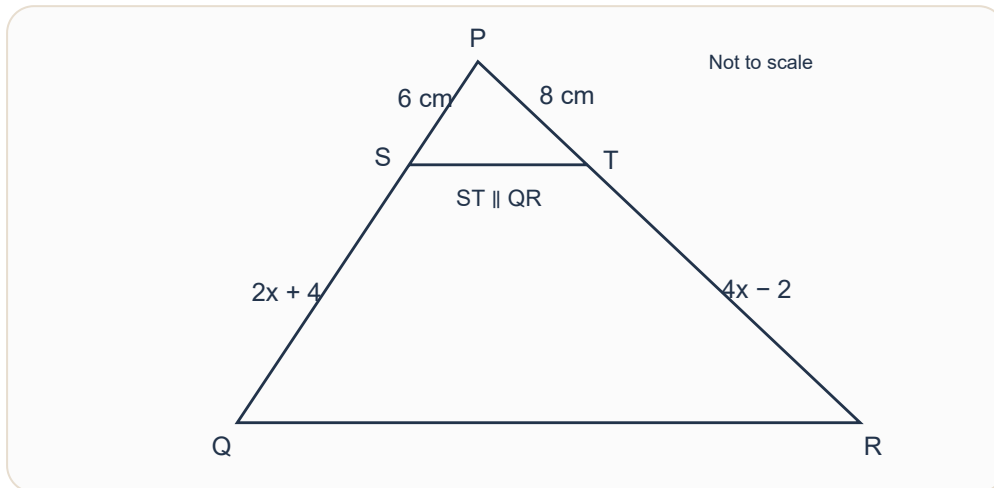
$$x^2 - 4x - 5 = 0 = (x - 5)(x + 1).$$

So $x = 5$. The original dimensions are 9 and 6, so the perimeter is $2(9 + 6) = \boxed{30 \text{ cm}}$.

Question 21

In triangle PQR , point S lies on PQ and point T lies on PR . The line ST is parallel to QR .

The lengths are labelled as follows: $PS = 6 \text{ cm}$, $SQ = (2x + 4) \text{ cm}$, $PT = 8 \text{ cm}$, $TR = (4x - 2) \text{ cm}$.



Given that the area of triangle PST is 12 cm^2 , what is the area of trapezium $STQR$?

A 96 cm^2

B 120 cm^2

C 135 cm^2

D 144 cm^2

E 147 cm^2

WORKED SOLUTION

Similarity gives $PS/PQ = PT/PR$. Here $PQ = 6 + (2x + 4) = 2x + 10$ and $PR = 8 + (4x - 2) = 4x + 6$. Therefore

$$\frac{6}{2x + 10} = \frac{8}{4x + 6} \Rightarrow 24x + 36 = 16x + 80 \Rightarrow x = \frac{11}{2}.$$

Then $PQ = 21$ and $PR = 28$, so the small-to-large length ratio is $2/7$. The area ratio is $4/49$.

$$\text{Area}(PQR) = 12 \times \frac{49}{4} = 147 \text{ cm}^2.$$

Subtract the small triangle: $147 - 12 = \boxed{135 \text{ cm}^2}$. The stated small triangle is feasible: its area is at most $\frac{1}{2}(6)(8) = 24 \text{ cm}^2$.

Question 22

Charlie has a bowl containing red sweets and green sweets only. There are **9** more red sweets than green sweets.

Two sweets are chosen at random without replacement.

The probability that the two sweets are the same colour is $\frac{4}{7}$.

How many sweets are there in the bowl altogether?

A 18

B 21

C 24

D 27

E 30

WORKED SOLUTION

Let the number of green sweets be g . Then the number of red sweets is $g + 9$, so total sweets are $2g + 9$.

The probability condition is

$$\frac{\binom{g+9}{2} + \binom{g}{2}}{\binom{2g+9}{2}} = \frac{4}{7}.$$

This simplifies to

$$(g - 6)(g + 18) = 0.$$

So $g = 6$. Then red sweets = **15**, giving total sweets = **21**.

Question 23

The point $(-1, 5)$ is translated by two successive vectors.

The first translation is by $\begin{pmatrix} 3p \\ 2 - q \end{pmatrix}$, and the second is by $\begin{pmatrix} q + 1 \\ p - 3 \end{pmatrix}$.

The final image is $(8, 4)$.

What is the value of $p^2 + q^2$?

A 4

B 5

C 6

D 8

E 10

WORKED SOLUTION

The total translation vector is

$$\begin{pmatrix} 3p + q + 1 \\ p - q - 1 \end{pmatrix}.$$

Since $(-1, 5) \rightarrow (8, 4)$, the net translation is $\begin{pmatrix} 9 \\ -1 \end{pmatrix}$.

So

$$3p + q = 8, \quad p - q = 0.$$

Thus $p = q$, so $4p = 8 \Rightarrow p = 2$, hence $q = 2$.

Therefore $p^2 + q^2 = 4 + 4 = \boxed{8}$.

Question 24

Consider the family of graphs

$$y = ax^2 + 2(a - 1)x + a + 3.$$

What is the complete range of values of a for which the minimum point of the graph lies above the x -axis?

A $a > \frac{1}{5}$

B $a > 0$

C $a \geq \frac{1}{5}$

D $a < 0$

E $0 < a < \frac{1}{5}$

WORKED SOLUTION

For there to be a minimum, we need $a > 0$.

For $y = ax^2 + bx + c$, the minimum value is $c - \frac{b^2}{4a}$.

Here $b = 2(a - 1)$ and $c = a + 3$, so the minimum value is

$$a + 3 - \frac{(2(a - 1))^2}{4a} = a + 3 - \frac{(a - 1)^2}{a} = 5 - \frac{1}{a}.$$

We need this to be positive:

$$5 - \frac{1}{a} > 0.$$

Since $a > 0$, this gives $5a > 1 \Rightarrow a > \frac{1}{5}$.

Question 25

An online company sells storage containers. The following items are available:

Capacity of container Number available

2 litres	2
3 litres	3
5 litres	4

A customer chooses a selection of containers with total capacity **23** litres.

Different selections are distinguished only by how many containers of each size are chosen.

If all possible selections are equally likely, what is the probability that the chosen selection contains at least one 2-litre container?

A $\frac{1}{3}$

B $\frac{1}{2}$

C $\frac{3}{5}$

D $\frac{3}{4}$

E $\frac{2}{3}$

WORKED SOLUTION

Let a, b, c be the numbers of 2-, 3- and 5-litre containers. Then

$$2a + 3b + 5c = 23,$$

with $0 \leq a \leq 2, 0 \leq b \leq 3, 0 \leq c \leq 4$.

The valid selections are

- $(0, 1, 4)$
- $(1, 2, 3)$
- $(2, 3, 2)$

Two of the three contain at least one 2-litre container, so the probability is $\boxed{\frac{2}{3}}$.

Question 26

The space diagonal of a cuboid is $7\sqrt{13}$ cm.

A similar cuboid has volume 8 times the volume of the original cuboid.

What is the space diagonal of the larger cuboid?

A $7\sqrt{26}$ cm

B $14\sqrt{7}$ cm

C $14\sqrt{13}$ cm

D $21\sqrt{13}$ cm

E $28\sqrt{13}$ cm

WORKED SOLUTION

A volume scale factor of 8 means a linear scale factor of $\sqrt[3]{8} = 2$.

So the new diagonal is

$$2 \cdot 7\sqrt{13} = \boxed{14\sqrt{13} \text{ cm}}.$$

Question 27

Alex, Cameron and Sam are running a **400 m** race. They each run at a different constant speed.

When Alex finishes, Cameron has run **300 m** and Sam has run **240 m**.

Alex beats Sam by **$33\frac{1}{3}$** seconds.

What is Cameron's speed?

A 5 m s^{-1}

B 6 m s^{-1}

C 6.4 m s^{-1}

D 7 m s^{-1}

E 8 m s^{-1}

WORKED SOLUTION

Let Alex's speed be A , Cameron's speed be C , and Sam's speed be S .

When Alex finishes, the elapsed time is $\frac{400}{A}$. In that time, Cameron runs 300 m, so

$$C \cdot \frac{400}{A} = 300 \Rightarrow \frac{C}{A} = \frac{3}{4}.$$

Similarly, Sam runs 240 m, so

$$S \cdot \frac{400}{A} = 240 \Rightarrow \frac{S}{A} = \frac{3}{5}.$$

Thus $S = \frac{3}{5}A$.

Alex beats Sam by $\frac{100}{3}$ s, so

$$\frac{400}{S} - \frac{400}{A} = \frac{100}{3}.$$

Substitute $S = \frac{3}{5}A$:

$$\frac{400}{3A/5} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{2000}{3A} - \frac{400}{A} = \frac{100}{3} \Rightarrow \frac{800}{3A} = \frac{100}{3}.$$

So $A = 8 \text{ m s}^{-1}$. Hence

$$C = \frac{3}{4}A = 6 \text{ m s}^{-1}.$$

Therefore Cameron's speed is **6 m s^{-1}** .

Question 1

A resistor has a constant potential difference of **9.0 V** across it. A total charge of **180 C** passes through it in **4.00 min**.

The resistor is then left connected for a total of **6.00 min**. Assume all the energy dissipated warms a **0.250 kg** metal block of specific heat capacity **400 J kg⁻¹K⁻¹**, and that there are no losses.

What is the temperature rise of the block?

Assume the resistor has constant resistance, so the current remains constant.

A 12.0 K

B 18.0 K

C 21.6 K

D 24.3 K

E 27.0 K

WORKED SOLUTION

The current is

$$I = \frac{Q}{t} = \frac{180}{240} = 0.75 \text{ A.}$$

So the power is

$$P = VI = 9.0 \times 0.75 = 6.75 \text{ W.}$$

Over **6.00 min = 360 s**, the energy transferred is

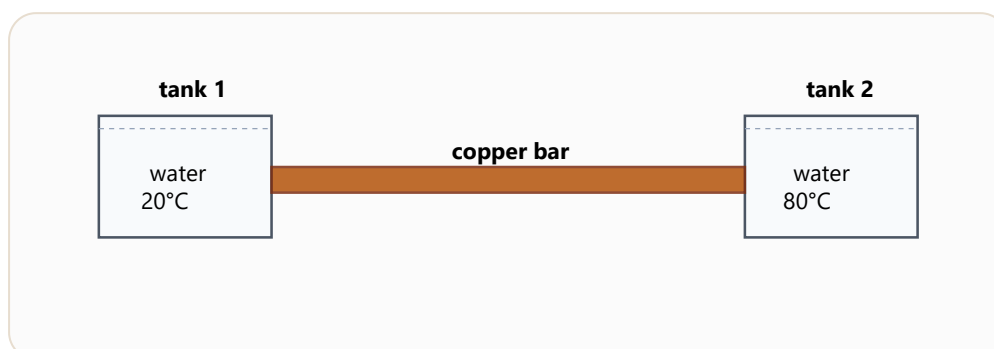
$$E = Pt = 6.75 \times 360 = 2430 \text{ J.}$$

Hence

$$\Delta T = \frac{E}{mc} = \frac{2430}{0.250 \times 400} = 24.3 \text{ K.}$$

Question 2

Two large copper tanks of water are connected by a solid cylindrical copper bar. Initially the water temperatures are 20°C and 80°C , and the bar has length L and diameter d .



The system is modified so that the temperatures become 35°C and 95°C , the bar length becomes $0.75L$, and the diameter becomes $1.5d$.

By what factor does the rate of conduction of thermal energy along the bar change?

Use $P = kA\Delta T/L$, where P is the rate of heat transfer, A is the cross-sectional area, ΔT is the temperature difference, and k is unchanged.

A 0.75

B 1.5

C 2.25

D 3.0

E 4.0

WORKED SOLUTION

For conduction,

$$P \propto \frac{A\Delta T}{L}.$$

The temperature difference stays 60°C , so ΔT is unchanged. The area factor is

$$\left(\frac{1.5d}{d}\right)^2 = 2.25.$$

The length factor is

$$\frac{L}{0.75L} = \frac{4}{3}.$$

Hence the overall factor is

$$2.25 \times \frac{4}{3} = 3.0.$$

Question 3

A car of mass **1400 kg** tows a caravan of mass **1000 kg** along a horizontal road at constant speed. The driving force from the car's engine is then increased by **3000 N**.

At one instant during the resulting acceleration, the resistive force on the car has increased by **200 N** and the resistive force on the caravan has increased by **400 N**, compared with their values at constant speed.

By how much has the tension in the tow bar increased compared with its value at constant speed?

A 600 N

B 1000 N

C 1400 N

D 1600 N

E 1800 N

WORKED SOLUTION

The initial resultant force is zero. Its increase is $3000 - 200 - 400 = 2400 \text{ N}$. Thus the acceleration is $a = 2400 / (1400 + 1000) = 1.0 \text{ m s}^{-2}$.

Let the initial caravan resistance and tow-bar tension be R_0 and T_0 . At constant speed, $T_0 = R_0$. During acceleration,

$$T - (R_0 + 400) = 1000(1.0).$$

Subtracting the initial balance gives $T - T_0 = 1000 + 400 = \boxed{1400 \text{ N}}$. This is the increase in tension.

Question 4

A small piece of space debris of mass **0.10 kg** strikes a surface of the International Space Station normally at a relative speed of $1.5 \times 10^4 \text{ m s}^{-1}$. The debris comes to rest relative to the station in **0.010 s**.

Assume the constant contact area during the impact is **2.0 cm²**.

What is the average pressure exerted on the station during the impact?

A $7.5 \times 10^6 \text{ Pa}$

B $7.5 \times 10^7 \text{ Pa}$

C $7.5 \times 10^8 \text{ Pa}$

D $1.5 \times 10^8 \text{ Pa}$

E $1.5 \times 10^9 \text{ Pa}$

WORKED SOLUTION

The average force magnitude is

$$F = \frac{\Delta p}{\Delta t} = \frac{0.10 \times 1.5 \times 10^4}{0.010} = 1.5 \times 10^5 \text{ N.}$$

The contact area is

$$2.0 \text{ cm}^2 = 2.0 \times 10^{-4} \text{ m}^2.$$

So the pressure is

$$p = \frac{F}{A} = \frac{1.5 \times 10^5}{2.0 \times 10^{-4}} = 7.5 \times 10^8 \text{ Pa.}$$

Question 5

A star is moving away from a space telescope. The star emits light of wavelength **500 nm**, but the telescope detects it with wavelength **502 nm**.

What is the approximate percentage decrease in the detected frequency, and hence in the energy of each detected photon?

Use $c = f\lambda$ and $E = hf$, where c is the speed of light, f is frequency, λ is wavelength, E is photon energy and h is a constant.

A 0.20%

B 0.40%

C 0.80%

D 1.0%

E 2.0%

WORKED SOLUTION

In vacuum, $c = f\lambda$, so $f \propto \frac{1}{\lambda}$.

The fractional decrease in frequency is therefore approximately the same as the fractional increase in wavelength:

$$\frac{502 - 500}{502} \approx \frac{2}{502} \approx 0.00398.$$

This is about **0.40%**. Since photon energy is $E = hf$, the percentage decrease in photon energy is the same.

Question 6

An electric motor is supplied with power at **0.800 kW**. Its efficiency is **60%**, and it runs for half an hour while lifting a load vertically at constant speed.

Ignoring any change in the motor itself, what maximum mass can be lifted through a height of **180 m**? Take $g = 10 \text{ N kg}^{-1}$.

A 120 kg**B** 240 kg**C** 360 kg**D** 480 kg**E** 720 kg**WORKED SOLUTION**

Input power is **800 W**, so useful power is

$$0.60 \times 800 = 480 \text{ W}.$$

Over **1800 s**, the useful energy is

$$E = 480 \times 1800 = 864000 \text{ J}.$$

This equals the gain in gravitational potential energy:

$$mgh = 864000.$$

Hence

$$m = \frac{864000}{10 \times 180} = 480 \text{ kg}.$$

Question 7

Air is trapped in a cylinder by a freely moving piston. Initially the trapped air has density ρ at temperature **300 K**.

The external load on the piston is adjusted so that the pressure of the trapped air becomes **20%** greater than its initial pressure. The load is then kept unchanged, so the pressure remains at this value while the trapped air cools to **270 K**. No gas escapes.

Assume the air behaves as an ideal gas. What is the final density of the trapped air?

For a fixed mass of ideal gas, pV/T is constant and $\rho = m/V$, where p is pressure, V is volume, T is absolute temperature and m is mass.

A 1.09ρ

B 1.20ρ

C 1.25ρ

D $\frac{4}{3}\rho$

E 1.44ρ

WORKED SOLUTION

For a fixed gas mass,

$$p = \rho RT \Rightarrow \rho \propto \frac{p}{T}.$$

So the density factor is

$$\frac{\rho_f}{\rho_i} = \frac{1.20}{270/300} = 1.20 \times \frac{300}{270} = \frac{4}{3}.$$

Therefore $\rho_f = \frac{4}{3}\rho$.

Question 8

A steady current flows in a conducting wire. A charge of **48 C** passes a point on the wire in **2.0 min**.

A straight **36 cm** section of the wire lies in a uniform magnetic field of strength **0.20 T**. This section makes an angle of **30°** to the direction of the field.

What is the magnitude of the magnetic force on this section?

Use $F = BIL \sin \theta$, where B is magnetic field strength, I is current, L is wire length in the field and θ is the angle between the wire and the field.

A $2.88 \times 10^{-3} \text{ N}$

B $7.2 \times 10^{-3} \text{ N}$

C $1.44 \times 10^{-2} \text{ N}$

D $2.88 \times 10^{-2} \text{ N}$

E $5.76 \times 10^{-2} \text{ N}$

WORKED SOLUTION

The current is

$$I = \frac{Q}{t} = \frac{48}{120} = 0.40 \text{ A.}$$

The magnetic force on a wire is

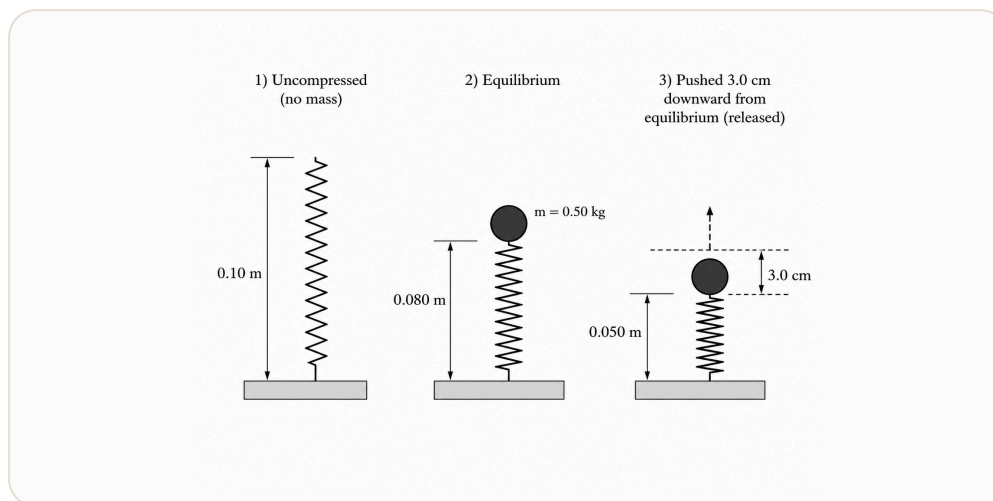
$$F = BIL \sin \theta.$$

So

$$F = 0.20 \times 0.40 \times 0.36 \times \sin 30^\circ = 0.20 \times 0.40 \times 0.36 \times 0.5 = 1.44 \times 10^{-2} \text{ N.}$$

Question 9

A light spring has uncompressed length **0.10 m**. When an object of mass **0.50 kg** rests on top of the spring in equilibrium, the spring length is **0.080 m**.



The mass is then pushed a further **3.0 cm** downward from equilibrium and released, moving on a frictionless vertical guide.

What is the speed of the mass as it passes back through its equilibrium position?

Take $g = 10 \text{ N kg}^{-1}$.

A 0.30 m s^{-1}

B 0.47 m s^{-1}

C 0.60 m s^{-1}

D 0.67 m s^{-1}

E 0.95 m s^{-1}

WORKED SOLUTION

At equilibrium, the spring is compressed by **0.020 m**, so

$$kx = mg \Rightarrow k = \frac{0.50 \times 10}{0.020} = 250 \text{ N m}^{-1}.$$

If the mass is displaced by an additional $a = 0.030 \text{ m}$ from equilibrium, then at the equilibrium position all of the extra elastic/gravitational energy has become kinetic energy:

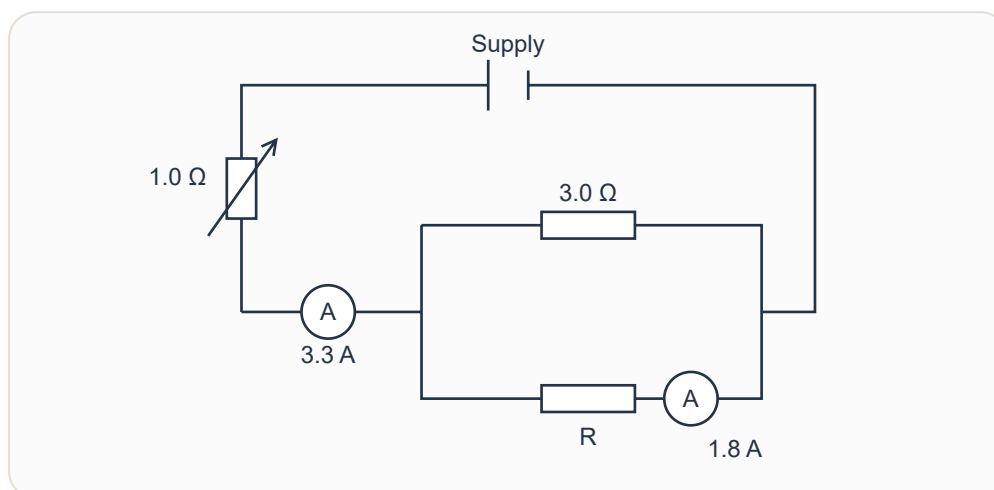
$$\frac{1}{2}ka^2 = \frac{1}{2}mv^2.$$

Hence

$$v = a\sqrt{\frac{k}{m}} = 0.030\sqrt{\frac{250}{0.50}} = 0.030\sqrt{500} \approx 0.67 \text{ m s}^{-1}.$$

Question 10

The circuit includes a variable resistor in series with a parallel network containing a $3.0\ \Omega$ resistor and a resistor R . The total current is $3.3\ \text{A}$, and the current through the R branch is $1.8\ \text{A}$.



The variable resistor has resistance $1.0\ \Omega$. Assume ideal ammeters and negligible internal resistance of the supply.

What is the emf of the supply?

A $4.5\ \text{V}$

B $6.6\ \text{V}$

C $7.8\ \text{V}$

D $9.0\ \text{V}$

E $12.3\ \text{V}$

WORKED SOLUTION

The right ammeter is in the R branch, so the current in R is $1.8\ \text{A}$.

The current in the $3.0\ \Omega$ branch is

$$3.3 - 1.8 = 1.5\ \text{A}.$$

So the p.d. across the parallel network is

$$V_{\parallel} = 1.5 \times 3.0 = 4.5\ \text{V}.$$

The current through the series variable resistor is the total current $3.3\ \text{A}$, so its p.d. is

$$V_{\text{var}} = 3.3 \times 1.0 = 3.3\ \text{V}.$$

Hence the supply emf is

$$\mathcal{E} = 4.5 + 3.3 = 7.8\ \text{V}.$$

Question 11

Element Q has several isotopes. Consider the two isotopes

$${}_{x}^{3x-7}Q \quad \text{and} \quad {}_{x}^{2x+3}Q.$$

The first contains 6 more neutrons than the second.

Another isotope of Q is

$${}_{x}^{3x+1}Q.$$

How many neutrons does this third isotope contain?

A 19

B 21

C 25

D 29

E 33

WORKED SOLUTION

The neutron numbers are:

$$\text{first isotope: } (3x - 7) - x = 2x - 7$$

$$\text{second isotope: } (2x + 3) - x = x + 3$$

Given that the first has 6 more neutrons,

$$(2x - 7) - (x + 3) = 6.$$

So

$$x - 10 = 6 \Rightarrow x = 16.$$

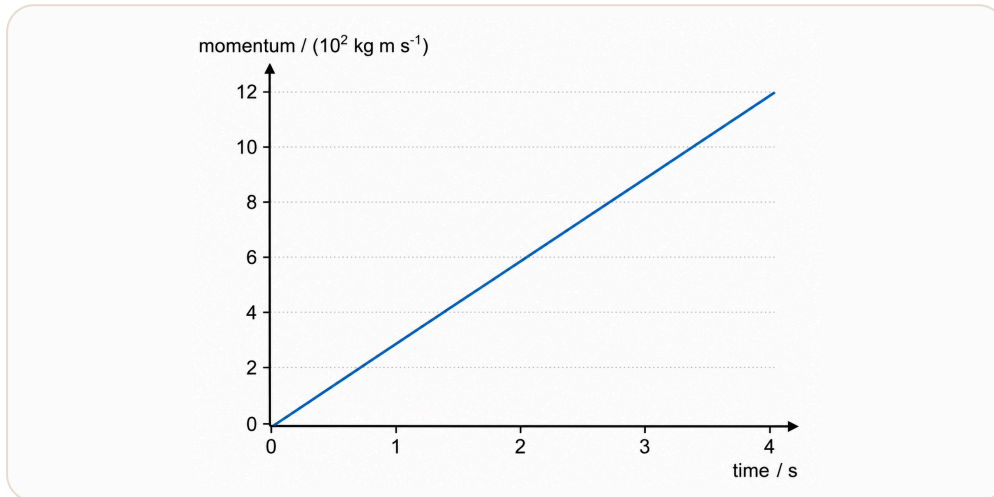
For the third isotope, the number of neutrons is

$$(3x + 1) - x = 2x + 1 = 2(16) + 1 = 33.$$

Question 12

A motorised trolley of mass $2.4 \times 10^3 \text{ kg}$ accelerates from rest along a horizontal track. The friction force is constant at $1.8 \times 10^3 \text{ N}$.

The graph shows how the momentum of the trolley varies with time.



What is the power delivered by the trolley's motor at $t = 4.0 \text{ s}$?

A $1.5 \times 10^2 \text{ W}$

B $3.0 \times 10^2 \text{ W}$

C $6.3 \times 10^2 \text{ W}$

D $1.05 \times 10^3 \text{ W}$

E $2.1 \times 10^3 \text{ W}$

WORKED SOLUTION

The graph rises from zero to $12 \times 10^2 = 1200 \text{ kg m s}^{-1}$ in 4.0 s . Its gradient is the resultant force:

$$F_{\text{net}} = 1200/4 = 300 \text{ N}.$$

The motor must also overcome friction, so $F_{\text{motor}} = 300 + 1800 = 2100 \text{ N}$.

At $t = 4.0 \text{ s}$, the speed is $v = p/m = 1200/2400 = 0.50 \text{ m s}^{-1}$.

$$P = F_{\text{motor}}v = 2100(0.50) = \boxed{1.05 \times 10^3 \text{ W}}.$$

Question 13

A block of mass **6.0 kg** is pushed along a rough horizontal surface by a constant force of **8.0 N**. It accelerates uniformly from rest and reaches speed **2.0 m s⁻¹** after **4.0 s**.

What fraction of the work done by the pushing force during this **4.0 s** is converted into thermal energy against resistive forces?

A $\frac{3}{8}$

B $\frac{1}{2}$

C $\frac{5}{8}$

D $\frac{3}{4}$

E $\frac{5}{6}$

WORKED SOLUTION

The acceleration is

$$a = \frac{2.0}{4.0} = 0.50 \text{ m s}^{-2}.$$

The resultant force is therefore

$$F_{\text{net}} = ma = 6.0 \times 0.50 = 3.0 \text{ N}.$$

So the resistive force is

$$8.0 - 3.0 = 5.0 \text{ N}.$$

The displacement in **4.0 s** is

$$s = \frac{1}{2}at^2 = \frac{1}{2}(0.50)(4.0)^2 = 4.0 \text{ m}.$$

Work done by the pushing force:

$$W_{\text{push}} = 8.0 \times 4.0 = 32 \text{ J}.$$

Thermal energy against resistance:

$$W_{\text{res}} = 5.0 \times 4.0 = 20 \text{ J}.$$

Hence the fraction is

$$\frac{20}{32} = \frac{5}{8}.$$

Question 14

A metal block of mass M is heated at a constant rate P . The graph of temperature rise ΔT against time t is a straight line of gradient k .

A second block is made of the same material, has mass $3M$, and is heated at constant rate $2P$.

What is the gradient of its ΔT -against- t graph?

Assume no thermal energy is lost to the surroundings.

A $\frac{k}{6}$

B $\frac{k}{3}$

C $\frac{2k}{3}$

D $\frac{3k}{2}$

E $6k$

WORKED SOLUTION

With no heat loss, $Pt = Mc\Delta T$. The straight-line gradient is therefore $k = \Delta T/t = P/(Mc)$.

For the second block,

$$k_2 = \frac{2P}{3Mc} = \frac{2k}{3}.$$

Question 15

A non-ideal transformer has **100** turns on the primary coil and **25** turns on the secondary coil. It is supplied with **3.0 kW** of electrical power at a current of **12.5 A**.

The output voltage is the same as for an ideal transformer, but the current in the secondary coil is **40 A**.

The transformer runs for **15 min**. Assume all power lost by inefficiency remains in the transformer, whose thermal capacity is $2.0 \times 10^4 \text{ J K}^{-1}$.

What is the temperature rise of the transformer?

A 12 K

B 18 K

C 24 K

D 27 K

E 30 K

WORKED SOLUTION

Input voltage is

$$V_p = \frac{P_{in}}{I_p} = \frac{3000}{12.5} = 240 \text{ V.}$$

For the turns ratio,

$$\frac{V_s}{V_p} = \frac{25}{100} = \frac{1}{4},$$

so $V_s = 60 \text{ V}$.

Thus the output power is

$$P_{out} = V_s I_s = 60 \times 40 = 2400 \text{ W.}$$

Power lost is

$$P_{loss} = 3000 - 2400 = 600 \text{ W.}$$

In **900 s**, the thermal energy gained is

$$E = 600 \times 900 = 5.4 \times 10^5 \text{ J.}$$

So the temperature rise is

$$\Delta T = \frac{E}{C} = \frac{5.4 \times 10^5}{2.0 \times 10^4} = 27 \text{ K.}$$

Question 16

A set of decorative lights consists of **20** identical lamps connected in series to a d.c. supply of constant voltage. The total power transferred by all **20** lamps together is P .

One lamp fails so that it becomes short-circuited and has zero resistance. The remaining **19** lamps are still lit.

By what factor does the power transferred to *each remaining lamp* increase?

Assume every functioning lamp has the same constant resistance before and after the failure.

A $\frac{19}{20}$

B $\frac{20}{19}$

C $\frac{361}{400}$

D $\frac{400}{361}$

E $\frac{400}{19}$

WORKED SOLUTION

If each lamp has resistance R , then the original total resistance is $20R$, so current is

$$I = \frac{V}{20R}.$$

After one lamp is short-circuited, the total resistance is $19R$, so current becomes

$$I' = \frac{V}{19R}.$$

Power in each functioning lamp is proportional to $I^2 R$. Therefore the factor increase is

$$\left(\frac{I'}{I}\right)^2 = \left(\frac{V/(19R)}{V/(20R)}\right)^2 = \left(\frac{20}{19}\right)^2 = \frac{400}{361}.$$

Question 17

The wavelength range of visible light is about **400 nm** to **700 nm**. Light of frequency $6.0 \times 10^{14} \text{ Hz}$ is green. Microwaves used in cooking have wavelength **12 cm**.

Which statements are correct?

1. Light of frequency $7.5 \times 10^{14} \text{ Hz}$ is red.
2. Microwaves used in cooking have frequency $2.5 \times 10^9 \text{ Hz}$.
3. Electromagnetic radiation of frequency $3.0 \times 10^{13} \text{ Hz}$ can be used for thermal imaging of buildings.

Take $c = 3.0 \times 10^8 \text{ m s}^{-1}$.

A none of them

B 1 only

C 2 only

D 3 only

E 2 and 3 only

WORKED SOLUTION

Statement 1 is false: a higher visible frequency corresponds to bluer/violet light, not red.

For statement 2,

$$f = \frac{c}{\lambda} = \frac{3.0 \times 10^8}{0.12} = 2.5 \times 10^9 \text{ Hz},$$

so statement 2 is true.

Statement 3 is also true: $3.0 \times 10^{13} \text{ Hz}$ lies in the infrared region, which is used in thermal imaging.

Question 18

A 6.0 V battery is connected in series with an 8.0Ω resistor, a filament lamp and an ammeter. The ammeter reading is 0.25 A .

Which row gives the potential difference, resistance and power of the lamp, in that order, at this operating point? Assume an ideal battery and ammeter.

A 2.0 V , 8.0Ω , 0.50 W

B 4.0 V , 8.0Ω , 1.0 W

C 4.0 V , 16Ω , 0.50 W

D 4.0 V , 16Ω , 1.0 W

E 6.0 V , 24Ω , 1.5 W

WORKED SOLUTION

The resistor has p.d.

$$V_R = IR = 0.25 \times 8.0 = 2.0 \text{ V}.$$

So the lamp has

$$V_L = 6.0 - 2.0 = 4.0 \text{ V}.$$

Its resistance at this operating point is

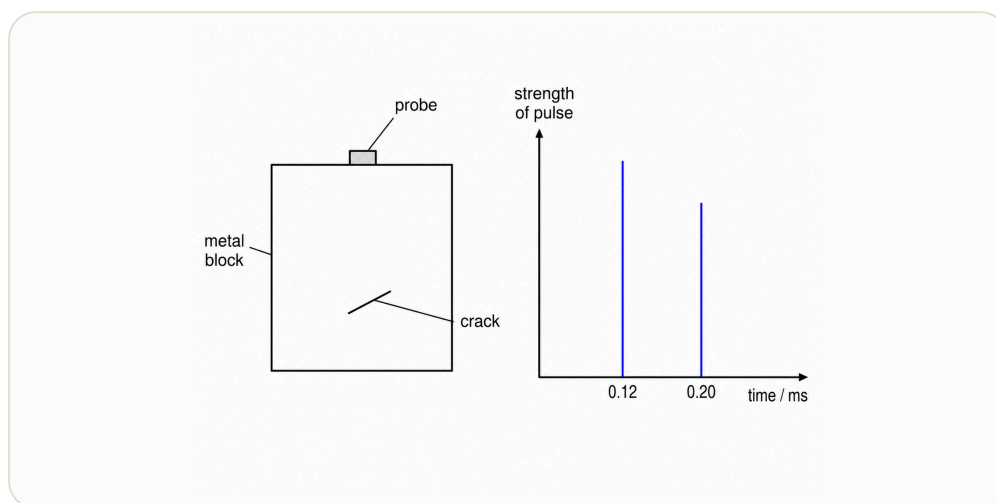
$$R_L = \frac{V_L}{I} = \frac{4.0}{0.25} = 16 \Omega.$$

Its power is

$$P_L = V_L I = 4.0 \times 0.25 = 1.0 \text{ W}.$$

Question 19

Ultrasound is used to find a crack inside a cuboid metal block. A probe is held against the top surface and sends a short pulse into the block. Reflections return from the crack and from the bottom surface.



The reflections return after **0.12 ms** and **0.20 ms**. The speed of ultrasound in the metal is **5000 m s⁻¹**. What is the ratio

$\frac{\text{thickness of block}}{\text{depth of crack below the top surface}}?$

A 2 : 1

B 3 : 2

C 5 : 3

D 4 : 3

E 5 : 4

WORKED SOLUTION

Echo times are round-trip times, so

depth of crack:

$$d = \frac{vt}{2} = \frac{5000 \times 0.12 \times 10^{-3}}{2} = 0.30 \text{ m.}$$

Thickness of block:

$$h = \frac{5000 \times 0.20 \times 10^{-3}}{2} = 0.50 \text{ m.}$$

Therefore

$$h : d = 0.50 : 0.30 = 5 : 3.$$

Question 20

A straight railway line between two cities is **60 km** long. A train starts from rest, accelerates uniformly at **1.5 m s^{-2}** to a maximum speed of **120 m s^{-1}** , then later brakes uniformly at **2.0 m s^{-2}** to come to rest at the second city.

What is the minimum possible journey time?

A 430 s

B 500 s

C 570 s

D 640 s

E 860 s

WORKED SOLUTION

Acceleration phase:

$$t_1 = \frac{120}{1.5} = 80 \text{ s}, \quad s_1 = \frac{120^2}{2(1.5)} = 4800 \text{ m.}$$

Braking phase:

$$t_2 = \frac{120}{2.0} = 60 \text{ s}, \quad s_2 = \frac{120^2}{2(2.0)} = 3600 \text{ m.}$$

Total distance used in speeding up and slowing down:

$$4800 + 3600 = 8400 \text{ m.}$$

Remaining cruising distance:

$$60000 - 8400 = 51600 \text{ m.}$$

Cruising time:

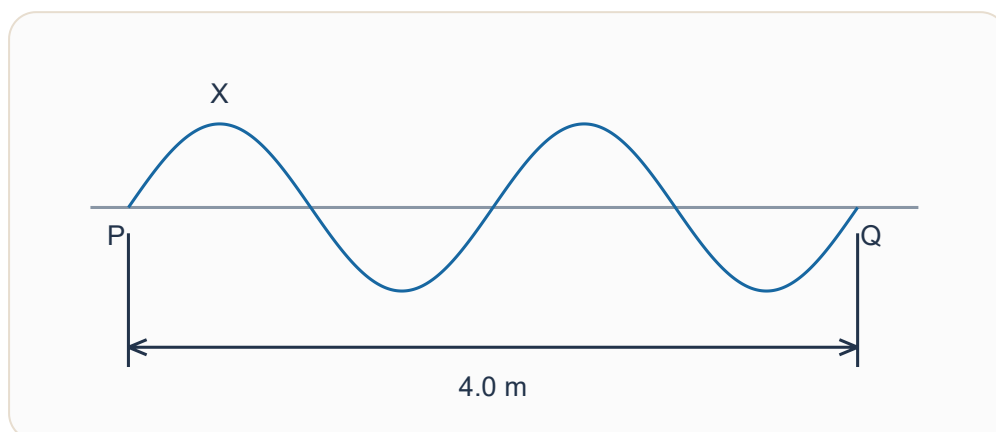
$$t_3 = \frac{51600}{120} = 430 \text{ s.}$$

Total time:

$$80 + 60 + 430 = 570 \text{ s.}$$

Question 21

A transverse wave on a string has speed 500 m s^{-1} . The horizontal distance between points P and Q shown is 4.0 m .



At time $t = 0$, point X is at its maximum displacement of 6.0 mm above equilibrium.

What is the displacement of point X at $t = 7.0 \text{ ms}$, and in which direction is it moving at that instant?

A 6.0 mm above equilibrium, moving downward

B 0 mm , moving upward

C 0 mm , moving downward

D 6.0 mm below equilibrium, moving upward

E between 0 and 6.0 mm below equilibrium, moving downward

WORKED SOLUTION

From the diagram, P to Q spans two wavelengths, so

$$2\lambda = 4.0 \text{ m} \Rightarrow \lambda = 2.0 \text{ m}.$$

Hence the frequency is

$$f = \frac{v}{\lambda} = \frac{500}{2.0} = 250 \text{ Hz}.$$

Therefore the period is

$$T = \frac{1}{f} = 0.004 \text{ s} = 4.0 \text{ ms}.$$

Now $7.0 \text{ ms} = 1.75T$, which is equivalent to $0.75T$ after the start. Starting at maximum displacement, after $0.75T$ the point is at equilibrium and moving upward.

Question 22

A radioactive nuclide X decays in a single stage to a stable nuclide R . Another radioactive nuclide Y decays in a single stage to a stable nuclide S .

When a rock formed, it contained equal numbers of atoms of X , Y , R and S .

The half-life of X is T years and the half-life of Y is $2T$ years.

What is the value of

$\frac{\text{number of atoms of } R}{\text{number of atoms of } S}$ at a time $4T$ years after the rock formed?

A $\frac{17}{20}$

B $\frac{31}{28}$

C $\frac{5}{4}$

D $\frac{6}{5}$

E 2

WORKED SOLUTION

Let each initial number be N .

For X : after $4T$, four half-lives have passed, so

$$X = \frac{N}{16}.$$

Thus $\frac{15N}{16}$ atoms have decayed into R , giving

$$R = N + \frac{15N}{16} = \frac{31N}{16}.$$

For Y : after $4T$, two half-lives have passed, so

$$Y = \frac{N}{4}.$$

Hence $\frac{3N}{4}$ atoms have decayed into S , giving

$$S = N + \frac{3N}{4} = \frac{7N}{4}.$$

Therefore

$$\frac{R}{S} = \frac{31N/16}{7N/4} = \frac{31}{28}.$$

Question 23

A ship travels into a wave that is moving in the opposite direction to the ship. The ship has speed 8.0 m s^{-1} and the wave has speed 3.0 m s^{-1} .

The front of the ship rises and falls with a time period of 8.0 s .

What are the wavelength of the wave and the period of the wave for a stationary buoy?

A 24 m, 8.0 s

B 40 m, 13.3 s

C 64 m, 21.3 s

D 88 m, 29.3 s

E 88 m, 8.0 s

WORKED SOLUTION

Relative speed between ship and wave crests is

$$8.0 + 3.0 = 11.0 \text{ m s}^{-1}.$$

So the wavelength is

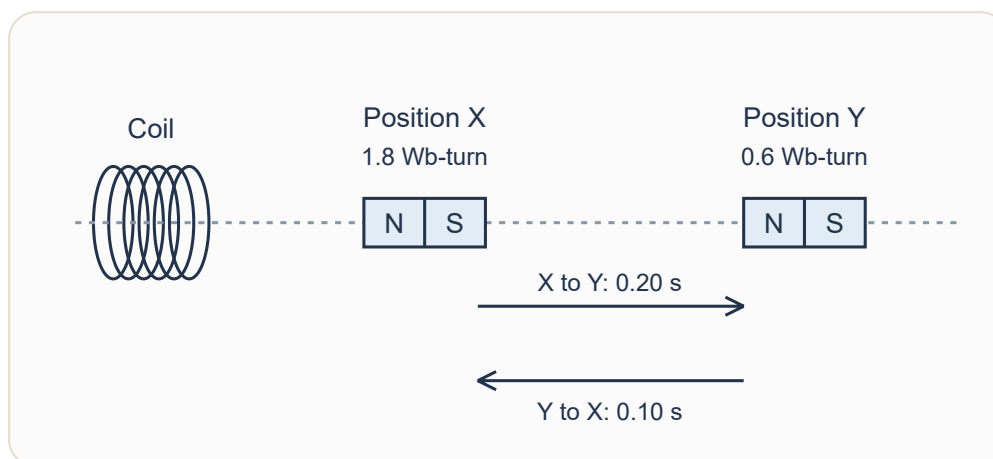
$$\lambda = v_{rel} T_{ship} = 11.0 \times 8.0 = 88 \text{ m}.$$

For a stationary buoy, the wave period is

$$T = \frac{\lambda}{v_{wave}} = \frac{88}{3.0} = 29.3 \text{ s}.$$

Question 24

A bar magnet is placed at position **X** close to one end of a coil and on its axis. It is moved rapidly to a more distant position **Y**, and then back to **X**.



The flux linkage of the coil is **1.8 Wb-turn** when the magnet is at **X** and **0.6 Wb-turn** when it is at **Y**. The motion from **X** to **Y** takes **0.20 s**, while the return motion from **Y** to **X** takes **0.10 s**.

Which is the larger of the magnitudes of the average induced e.m.f. for the two separate journeys?

Use $|\mathcal{E}_{\text{avg}}| = |\Delta(N\Phi)|/\Delta t$, where $N\Phi$ is the flux linkage and Δt is the time for one journey.

A 3.0 V

B 6.0 V

C 9.0 V

D 12 V

E 18 V

WORKED SOLUTION

The magnitude of the flux-linkage change is $1.8 - 0.6 = 1.2$ Wb-turn for each journey.

Outward: $|\mathcal{E}_{\text{avg}}| = 1.2/0.20 = 6.0$ V.

Return: $|\mathcal{E}_{\text{avg}}| = 1.2/0.10 = 12$ V.

The larger magnitude is **12 V**, on the return journey. The peak instantaneous e.m.f. cannot be determined from the two journey times alone.

Question 25

When an earthquake occurs, both P-waves and S-waves are produced at the same time and travel along the same path. P-waves travel at 5.0 km s^{-1} and S-waves travel at 3.0 km s^{-1} .

A seismic monitoring station detects the P-waves **30 s** before it detects the S-waves.

How long after the earthquake does the P-wave reach the station?

A 30 s

B 45 s

C 60 s

D 75 s

E 90 s

WORKED SOLUTION

Let the distance to the station be d km. Then

$$\frac{d}{3.0} - \frac{d}{5.0} = 30.$$

So

$$d \left(\frac{1}{3} - \frac{1}{5} \right) = 30 \Rightarrow d \left(\frac{2}{15} \right) = 30 \Rightarrow d = 225 \text{ km}.$$

The P-wave travel time is therefore

$$t_P = \frac{225}{5.0} = 45 \text{ s}.$$

Question 26

A tall smooth cylinder contains air at atmospheric pressure $1.00 \times 10^5 \text{ Pa}$. The density of the air is 1.20 kg m^{-3} . A heavy piston of mass 50.0 kg and cross-sectional area 0.0200 m^2 is placed on top and allowed to fall slowly until it rests in equilibrium.

The initial height of the trapped air column is 0.50 m . Assume the air behaves as an ideal gas and its temperature remains constant. Take $g = 10 \text{ N kg}^{-1}$.

By how far does the piston fall?

A 0.040 m

B 0.080 m

C 0.10 m

D 0.125 m

E 0.20 m

WORKED SOLUTION

The piston adds pressure

$$\frac{mg}{A} = \frac{50.0 \times 10}{0.0200} = 2.5 \times 10^4 \text{ Pa.}$$

So the final pressure is

$$1.00 \times 10^5 + 2.5 \times 10^4 = 1.25 \times 10^5 \text{ Pa.}$$

At constant temperature, pV is constant, so volume decreases by a factor

$$\frac{V_f}{V_i} = \frac{p_i}{p_f} = \frac{1.00}{1.25} = 0.80.$$

The cross-sectional area is constant, so the height also decreases by a factor of 0.80:

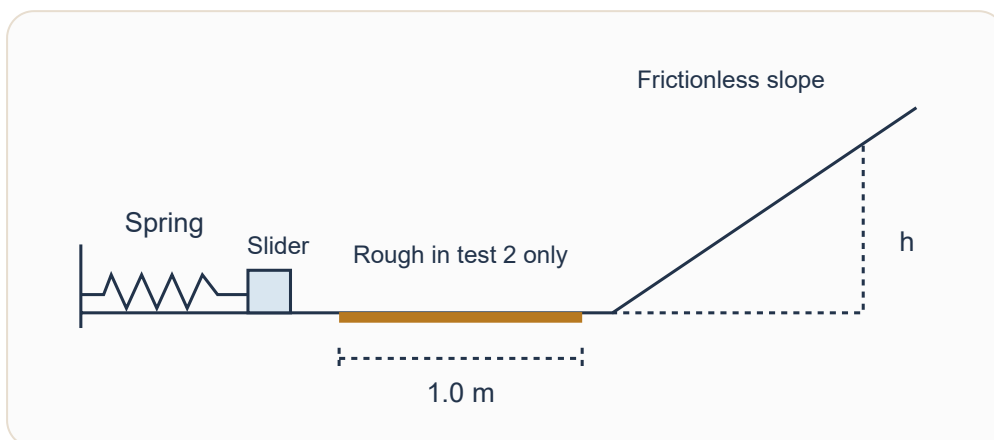
$$h_f = 0.80 \times 0.50 = 0.40 \text{ m.}$$

Therefore the piston falls by

$$0.50 - 0.40 = 0.10 \text{ m.}$$

Question 27

A light spring launches a slider horizontally towards a slope. In the first test, the horizontal track and the slope are both frictionless. The slider has mass **30 g**.



In one test, compressing the spring by **5.0 cm** sends the **30 g** slider up the slope to a maximum vertical height of **20 cm**.

The slider is then replaced by a **20 g** slider. The spring is compressed by **15 cm**. For this second test only, after leaving the spring the slider crosses a rough horizontal patch of length **1.0 m** on which a constant resistive force of **0.20 N** acts. Elsewhere resistance is negligible.

To what maximum vertical height above the horizontal launch track does the new slider rise? Take $g = 10 \text{ N kg}^{-1}$.

A 120 cm

B 150 cm

C 170 cm

D 180 cm

E 270 cm

WORKED SOLUTION

From the first test, spring energy equals gain in gravitational potential energy:

$$\frac{1}{2}k(0.050)^2 = 0.030 \times 10 \times 0.20.$$

So

$$0.00125k = 0.060 \Rightarrow k = 48 \text{ N m}^{-1}.$$

With the new slider and compression **0.15 m**, spring energy is

$$E_s = \frac{1}{2} \times 48 \times (0.15)^2 = 0.54 \text{ J}.$$

Work done against the rough patch is

$$W_f = 0.20 \times 1.0 = 0.20 \text{ J}.$$

So energy left for gravitational potential energy is

$$0.54 - 0.20 = 0.34 \text{ J}.$$

For the **20 g = 0.020 kg** slider,

$$mgh = 0.34 \Rightarrow h = \frac{0.34}{0.020 \times 10} = 1.7 \text{ m} = 170 \text{ cm}.$$

Question 1

The polynomial $P(x) = x^4 + ax^3 + bx^2 - 44x + 24$ has $(x - 1)$ and $(x - 2)$ as factors. What is the value of $a^2 + b$?

A 49**B** 61**C** 68**D** 75**E** 82**WORKED SOLUTION**

Since $x = 1$ and $x = 2$ are roots, $P(1)=0$ and $P(2)=0$.

From $P(1)=0$: $1 + a + b - 44 + 24 = 0$, so $a + b = 19$.

From $P(2)=0$: $16 + 8a + 4b - 88 + 24 = 0$, so $2a + b = 12$.

Subtracting gives $a = -7$, and then $b = 26$.

Hence $a^2 + b = 49 + 26 = \mathbf{75}$.

Question 2

The curve $y = x^3 + px^2 + qx + 10$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. What is the sum of the y -coordinates of these two stationary points?

A -8

B -2

C 2

D 6

E 12

WORKED SOLUTION

Differentiate: $y' = 3x^2 + 2px + q$. Since the stationary points are at $x = -1$ and $x = 3$, we have $y' = 3(x+1)(x-3) = 3x^2 - 6x - 9$.

So $2p = -6$ and $q = -9$, hence $p = -3$.

The curve is $y = x^3 - 3x^2 - 9x + 10$.

At $x = -1$, $y = -1 - 3 + 9 + 10 = 15$.

At $x = 3$, $y = 27 - 27 - 27 + 10 = -17$.

The sum is $15 + (-17) = -2$.

Question 3

The line $y = mx + 3$ is reflected in the line $y = x$. The reflected line has y-intercept $-\frac{2}{3}$. What is the value of m ?

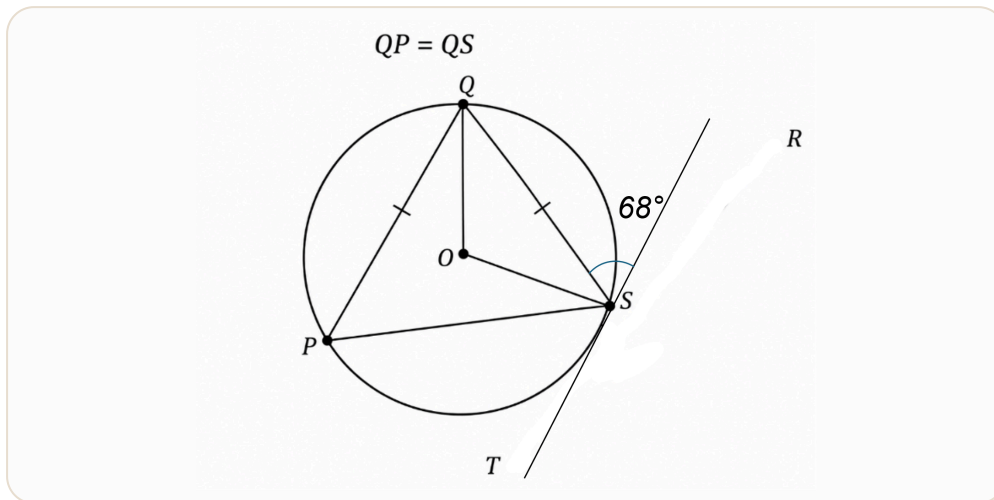
A $\frac{3}{2}$ **B** 2**C** $\frac{5}{2}$ **D** 3**E** $\frac{9}{2}$ **WORKED SOLUTION**

Reflecting in $y = x$ swaps x and y . So the reflected line is $x = my + 3$, which rearranges to $y = (x - 3)/m = (1/m)x - 3/m$.

The y-intercept of the reflected line is therefore $-3/m$.

Given $-3/m = -2/3$, we get $3/m = 2/3$, so $9 = 2m$ and $m = \mathbf{9/2}$.

Question 4



In the diagram, O is the centre of the circle and RT is a tangent at S . Points P and Q lie on the circumference, $QP = QS$, and the marked angle is $\angle QSR = 68^\circ$. What is the size of $\angle POS$?

A 44° B 68° C 88° D 112° E 136°

WORKED SOLUTION

By the tangent-chord theorem, $\angle QPS = \angle QSR = 68^\circ$.

Since $QP = QS$, the base angles at P and S are equal. Hence

$$\angle PQS = 180^\circ - 68^\circ - 68^\circ = 44^\circ.$$

The smaller central angle subtended by chord PS is twice the corresponding angle at the circumference:

$$\angle POS = 2(44^\circ) = \boxed{88^\circ}.$$

Question 5

What is the total area enclosed between the curve $y = x^2 - 4x + 3$ and the line $y = -x + 1$, for $0 \leq x \leq 3$?

A $\frac{7}{6}$ B $\frac{11}{6}$ C $\frac{13}{6}$ D $\frac{17}{6}$ E $\frac{23}{6}$

WORKED SOLUTION

Subtract the line from the curve: $(x^2 - 4x + 3) - (-x + 1) = x^2 - 3x + 2 = (x-1)(x-2)$.

This is positive on $[0,1]$, negative on $[1,2]$, and positive on $[2,3]$.

So total area = $\int_0^1 (x^2 - 3x + 2) dx - \int_1^2 (x^2 - 3x + 2) dx + \int_2^3 (x^2 - 3x + 2) dx$.

An antiderivative is $\frac{x^3}{3} - \frac{3x^2}{2} + 2x$.

Evaluating gives $\frac{5}{6} + \frac{1}{6} + \frac{5}{6} = \frac{11}{6}$.

Question 6

Simplify $\sqrt{18 + 8\sqrt{5}} - \sqrt{18 - 8\sqrt{5}}$. Write the answer in the form $k\sqrt{2}$. What is k ?

A 2**B** 3**C** 4**D** 5**E** 6**WORKED SOLUTION**

Write $18 + 8\sqrt{5}$ as $a + b + 2\sqrt{ab}$ with $a + b = 18$ and $ab = 80$. The numbers are 10 and 8.

So $\sqrt{18 + 8\sqrt{5}} = \sqrt{10 + \sqrt{8}} = \sqrt{10 + 2\sqrt{2}}$.

Similarly, $\sqrt{18 - 8\sqrt{5}} = \sqrt{10 - 2\sqrt{2}}$.

Subtracting gives $(\sqrt{10 + 2\sqrt{2}}) - (\sqrt{10 - 2\sqrt{2}}) = 4\sqrt{2}$.

Hence $k = 4$.

Question 7

The line $y = 2x + k$ is a tangent to the curve $y = 3x^2 - 10x + 7$. What is the value of k ?

A -7

B -5

C -3

D 1

E 5

WORKED SOLUTION

At the point of tangency the line and curve have one repeated intersection.

Set them equal: $3x^2 - 10x + 7 = 2x + k$, so $3x^2 - 12x + (7 - k) = 0$.

For tangency the discriminant is 0:

$$(-12)^2 - 4 \cdot 3 \cdot (7 - k) = 0.$$

$$144 - 12(7 - k) = 0 \Rightarrow 144 - 84 + 12k = 0 \Rightarrow 60 + 12k = 0.$$

So $k = -5$.

Question 8

Given $y = (3\sqrt{x} - 2/\sqrt{x})^2$, find dy/dx when $x = 1/2$.

A -9

B -7

C -5

D 7

E 9

WORKED SOLUTION

First simplify: $y = 9x - 12 + 4/x$.

Differentiate: $dy/dx = 9 - 4/x^2$.

At $x = 1/2$, $x^2 = 1/4$, so $4/x^2 = 16$.

Therefore $dy/dx = 9 - 16 = -7$.

Question 9

Find the value of $\int_1^4 (3x^2 + 1)/(x\sqrt{x}) \, dx$.

A 12

B 14

C 15

D 16

E 18

WORKED SOLUTION

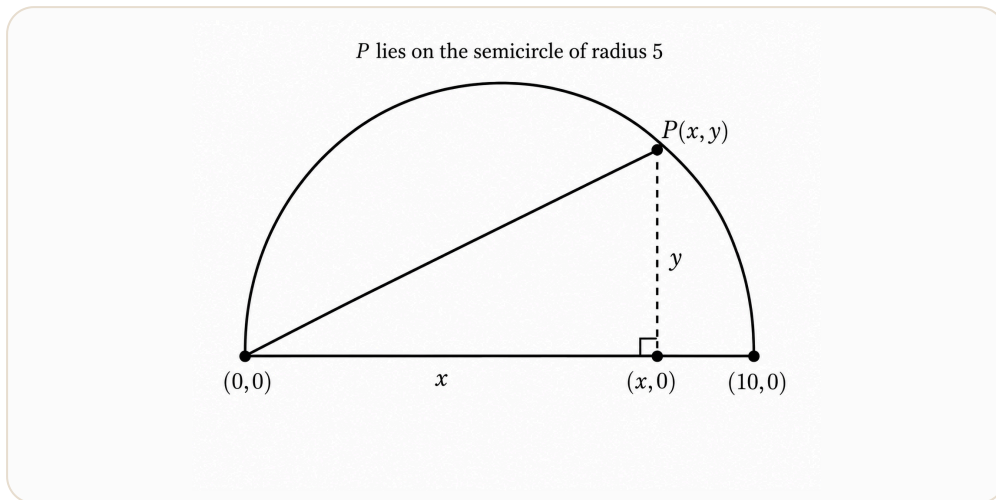
Rewrite the integrand: $(3x^2 + 1)/(x\sqrt{x}) = 3x^{1/2} + x^{-3/2}$.

So $\int (3x^{1/2} + x^{-3/2}) \, dx = 2x^{3/2} - 2x^{-1/2}$.

Evaluate from 1 to 4:

$$[2x^{3/2} - 2x^{-1/2}]_1^4 = (2 \cdot 8 - 2 \cdot 1/2) - (2 \cdot 1 - 2 \cdot 1) = 15 - 0 = \mathbf{15}.$$

Question 10



A point $P(x,y)$ lies on the semicircle with diameter from $(0,0)$ to $(10,0)$. A right triangle has vertices at $(0,0)$, $(x,0)$, and $P(x,y)$. What is the maximum possible area of the triangle?

A $25\sqrt{3/8}$

B $75\sqrt{3/8}$

C $25\sqrt{3/4}$

D $75\sqrt{3/4}$

E $125\sqrt{3/8}$

WORKED SOLUTION

The semicircle has centre $(5,0)$ and radius 5, so $y = \sqrt{25 - (x-5)^2} = \sqrt{10x - x^2}$.

The triangle area is $A = \frac{1}{2}xy = \frac{1}{2}x\sqrt{10x - x^2}$.

To maximise A , maximise $A^2 = \frac{1}{4}x^2(10x - x^2) = \frac{1}{4}(10x^3 - x^4)$.

Differentiate: $d(A^2)/dx = \frac{1}{4}(30x^2 - 4x^3) = x^2(15 - 2x)/2$.

So the maximum occurs at $x = 15/2$.

Then $y = \sqrt{10 \cdot 15/2 - (15/2)^2} = \sqrt{75/4} = 5\sqrt{3}/2$.

Hence $A = \frac{1}{2} \cdot 15/2 \cdot 5\sqrt{3}/2 = \mathbf{75\sqrt{3}/8}$.

Question 11

What is the complete set of real values of x for which $x^2(x^2 + 4) < 21$?

A $-2 < x < 2$

B $-\sqrt{3} < x < \sqrt{3}$

C $x < -\sqrt{3}$ or $x > \sqrt{3}$

D $-\sqrt{7} < x < \sqrt{7}$

E $-1 < x < 1$

WORKED SOLUTION

Let $u = x^2$, so $u \geq 0$. Then the inequality becomes $u(u+4) < 21$, i.e. $u^2 + 4u - 21 < 0$.

Factor: $(u+7)(u-3) < 0$, so $-7 < u < 3$.

Since $u = x^2 \geq 0$, we get $0 \leq u < 3$.

Thus $x^2 < 3$, so $-\sqrt{3} < x < \sqrt{3}$.

Question 12

Four mathematically similar solids W, X, Y and Z satisfy the following:

- the ratio of lengths W:X is 1:2
- the ratio of surface areas X:Y is 4:9
- the ratio of volumes Y:Z is 8:27

What is the ratio of the volumes of W:Z?

A 1:81

B 8:729

C 4:243

D 8:243

E 16:729

WORKED SOLUTION

Length ratio W:X = 1:2.

Surface area ratio X:Y = 4:9, so length ratio X:Y = 2:3.

Volume ratio Y:Z = 8:27, so length ratio Y:Z = 2:3.

Hence the overall length ratio W:Z = $1:2 \times 2:3 \times 2:3 = 2:9$.

Volumes scale as the cube of length, so W:Z volume ratio = $2^3:9^3 = \mathbf{8:729}$.

Question 13

What is the mean of $\log_2 64$, $\log_4 64$, and $\log_8 64$?

A 3

B $10/3$

C $11/3$

D 4

E $13/3$

WORKED SOLUTION

$\log_2 64 = 6$, $\log_4 64 = 3$, and $\log_8 64 = 2$.

So the mean is $(6 + 3 + 2)/3 = 11/3$.

Question 14

Which of the following is largest in value? (All angles are in radians.)

A $\sin 1$

B $\cos 0.5$

C $\tan 0.5$

D $\sin 0.5$

E $\cos 1$

WORKED SOLUTION

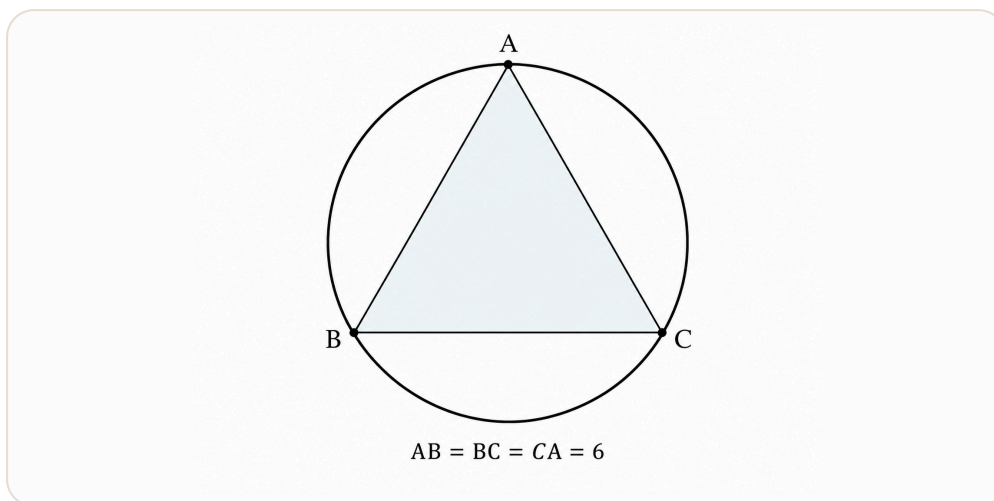
Since $\pi > 3$, $\pi/2 - 1 > 0.5$. Cosine decreases on $(0, \pi/2)$, so $\sin 1 = \cos(\pi/2 - 1) < \cos 0.5$. Also $\cos 1 < \cos 0.5$.

As $0.5 < \pi/6$,

$$\tan 0.5 < \tan(\pi/6) = \frac{1}{\sqrt{3}} < \frac{\sqrt{3}}{2} = \cos(\pi/6) < \cos 0.5.$$

Finally, $0.5 < \pi/4$ gives $\sin 0.5 < \cos 0.5$. Thus $\cos 0.5$ is largest, without numerical trigonometric approximations.

Question 15



An equilateral triangle of side 6 cm is inscribed in a circle. What is the area inside the circle but outside the triangle?

A $12\pi - 6\sqrt{3}$

B $12\pi - 9\sqrt{3}$

C $6\pi - 9\sqrt{3}$

D $4\pi - 9\sqrt{3}$

E $12\pi - 18\sqrt{3}$

WORKED SOLUTION

For an equilateral triangle of side s , the circumradius is $R = s/\sqrt{3}$.

So here $R = 6/\sqrt{3} = 2\sqrt{3}$.

Circle area $= \pi R^2 = \pi(2\sqrt{3})^2 = 12\pi$.

Triangle area $= (\sqrt{3}/4)s^2 = (\sqrt{3}/4) \cdot 36 = 9\sqrt{3}$.

Required area $= 12\pi - 9\sqrt{3}$, so the answer is **$12\pi - 9\sqrt{3}$** .

Question 16

An arithmetic sequence has first term a and common difference a . If S_n is the sum of the first n terms, which of the following is equal to $S_{2n} - 3S_n$?

A $an(n+1)/2$

B $an(2n-1)/2$

C $an(n-1)/2$

D $a(n-1)(n+1)$

E an^2

WORKED SOLUTION

The terms are $a, 2a, 3a, \dots, na$, so $S_n = a(1 + 2 + \dots + n) = an(n+1)/2$.

Then $S_{2n} = a \cdot 2n(2n+1)/2 = an(2n+1)$.

So $S_{2n} - 3S_n = an(2n+1) - 3[an(n+1)/2]$
 $= [2an(2n+1) - 3an(n+1)]/2 = an(n-1)/2$.

Thus the correct expression is **$an(n-1)/2$** .

Question 17

The equation $2 \cdot 5^{2x} - 7 \cdot 5^x + 3 = 0$ has two real solutions. What is the sum of these two solutions?

A $\log_5 3$

B $\log_5(1/2)$

C $\log_5(3/2)$

D $\log_5 6$

E $\log_5(2/3)$

WORKED SOLUTION

Let $u = 5^x$. Then $2u^2 - 7u + 3 = 0$, which factors as $(2u-1)(u-3)=0$.

So $u = 1/2$ or $u = 3$.

Hence $x = \log_5(1/2)$ or $x = \log_5 3$.

The sum is $\log_5(1/2) + \log_5 3 = \log_5(3/2)$.

Therefore the answer is **$\log_5(3/2)$** .

Question 18

The line L with equation $y = mx + c$, where $m > 0$ and $c \geq 0$, passes through the point $(2,4)$. A line is drawn through $(2,4)$ perpendicular to L . The triangle enclosed between the two lines and the y -axis has area 5 square units. What is the larger possible value of m ?

A $\frac{1}{2}$ **B** 1**C** $\frac{3}{2}$ **D** 2**E** $\frac{5}{2}$ **WORKED SOLUTION**

Since L passes through $(2,4)$, its equation is $y = mx + (4 - 2m)$. So its y -intercept is $4 - 2m$.

The perpendicular line through $(2,4)$ has slope $-1/m$, so its equation is $y - 4 = -(1/m)(x - 2)$. Its y -intercept is $4 + 2/m$.

The base of the triangle on the y -axis has length $(4 + 2/m) - (4 - 2m) = 2m + 2/m$.

The horizontal distance from $(2,4)$ to the y -axis is 2, so

Area = $\frac{1}{2} \times (2m + 2/m) \times 2 = 2(m + 1/m)$.

Set this equal to 5:

$$2(m + 1/m) = 5 \Rightarrow 2m^2 - 5m + 2 = 0 \Rightarrow (2m-1)(m-2)=0.$$

Thus $m = 1/2$ or 2, and the larger value is **2**.

Question 19

How many distinct real solutions does the equation $(x^2 - 2x - 3)^2 = x^2 - 2x - 3$ have?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let $u = x^2 - 2x - 3$. Then the equation becomes $u^2 = u$, so $u(u-1)=0$.

Hence $u = 0$ or $u = 1$.

If $u = 0$, then $x^2 - 2x - 3 = 0 \Rightarrow (x-3)(x+1)=0 \Rightarrow x = 3$ or -1 .

If $u = 1$, then $x^2 - 2x - 4 = 0 \Rightarrow x = 1 \pm \sqrt{5}$.

These are four distinct real solutions, so the answer is **4**.

Question 20

The curve $y = x^2 - x - 6$ intersects the x -axis at the points A and B, and has a minimum at the point C. A rectangle ABDE has two of its vertices at A and B, and C lies on the edge DE, between D and E. What is the area of the rectangle?

A 25

B $125/8$ C $125/4$

D 35

E 45

WORKED SOLUTION

The roots of $x^2 - x - 6 = 0$ are $x = 3$ and $x = -2$, so $AB = 5$.

The x -coordinate of the minimum is $x = -b/(2a) = 1/2$.

Then y at C is $(1/2)^2 - 1/2 - 6 = 1/4 - 1/2 - 6 = -25/4$.

Since C lies on the lower edge DE of the rectangle while A and B lie on the x -axis, the height of the rectangle is $25/4$.

Area = base $\times$ height = $5 \times 25/4 = 125/4$.

Question 21

It is given that $9\cos x + \tan x \cdot \sin x = 6$, where $0^\circ < x < 90^\circ$. What are the possible values of $\tan x$?

A $\frac{1}{2}$ or $\frac{1}{4}$

B $\frac{1}{3}$ or $\frac{1}{15}$

C $\sqrt{3}$ or $\sqrt{15}$

D 2 or 4

E $\sqrt{3}$ or $2\sqrt{2}$

WORKED SOLUTION

Let $c = \cos x$. Then $\tan x \cdot \sin x = \frac{\sin^2 x}{\cos x} = \frac{(1-c^2)}{c}$.

So the equation becomes $9c + \frac{(1-c^2)}{c} = 6$.

Multiply by c : $9c^2 + 1 - c^2 = 6c$, so $8c^2 - 6c + 1 = 0$.

Factor: $(4c-1)(2c-1)=0$, giving $c = \frac{1}{4}$ or $c = \frac{1}{2}$.

If $\cos x = \frac{1}{2}$, then $\tan x = \sqrt{3}$.

If $\cos x = \frac{1}{4}$, then $\sin x = \sqrt{1-\frac{1}{16}} = \frac{\sqrt{15}}{4}$, so $\tan x = \frac{(\sqrt{15}/4)}{(1/4)} = \sqrt{15}$.

Therefore the possible values are **$\sqrt{3}$ or $\sqrt{15}$** .

Question 22

A straight line passes through the points $(0, 3a)$ and $(2a, 0)$, where a is a positive constant. What is the perpendicular distance of the point $P(a, 4a)$ from this line?

A $\frac{6a}{\sqrt{13}}$

B $5\sqrt{13} a/13$

C $2\sqrt{13} a/13$

D $4\sqrt{13} a/13$

E $3\sqrt{13} a/13$

WORKED SOLUTION

The line through $(0, 3a)$ and $(2a, 0)$ has intercept form $x/(2a) + y/(3a) = 1$, so $3x + 2y - 6a = 0$.

The perpendicular distance from $(a, 4a)$ to this line is

$$|3a + 2(4a) - 6a| / \sqrt{3^2 + 2^2} = |5a|/\sqrt{13} = 5a/\sqrt{13} = \mathbf{5\sqrt{13} a/13}.$$

Question 23

At how many distinct points do the curves $y = (x - 3)(x^2 - 4x - 5)$ and $y = -x^2 + 6x - 9$ meet?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Set the equations equal:

$$(x - 3)(x^2 - 4x - 5) = -x^2 + 6x - 9.$$

Bring everything to one side:

$$(x - 3)(x^2 - 4x - 5) + x^2 - 6x + 9 = 0.$$

Since $x^2 - 6x + 9 = (x - 3)^2$, factor out $(x - 3)$:

$$(x - 3)[x^2 - 4x - 5 + (x - 3)] = (x - 3)(x^2 - 3x - 8) = 0.$$

So $x = 3$, or $x = (3 \pm \sqrt{41})/2$.

These are three distinct x -values, so the curves meet at **3** distinct points.

Question 24

A geometric progression has first term $u_1 = a$ and common ratio r . The sum to infinity is $\frac{8}{5}$, and the sum to infinity of the even-numbered terms ($u_2 + u_4 + u_6 + \dots$) is $\frac{3}{5}$. What is the value of $a + r$?

A $\frac{21}{25}$

B $\frac{6}{5}$

C $\frac{31}{25}$

D $\frac{8}{5}$

E $\frac{41}{25}$

WORKED SOLUTION

For the whole GP, $a/(1-r) = 8/5$.

The even-numbered terms form a GP with first term ar and ratio r^2 , so $ar/(1-r^2) = 3/5$.

But $a = (8/5)(1-r)$. Substitute into the second equation:

$$[(8/5)(1-r)r] / [(1-r)(1+r)] = 3/5.$$

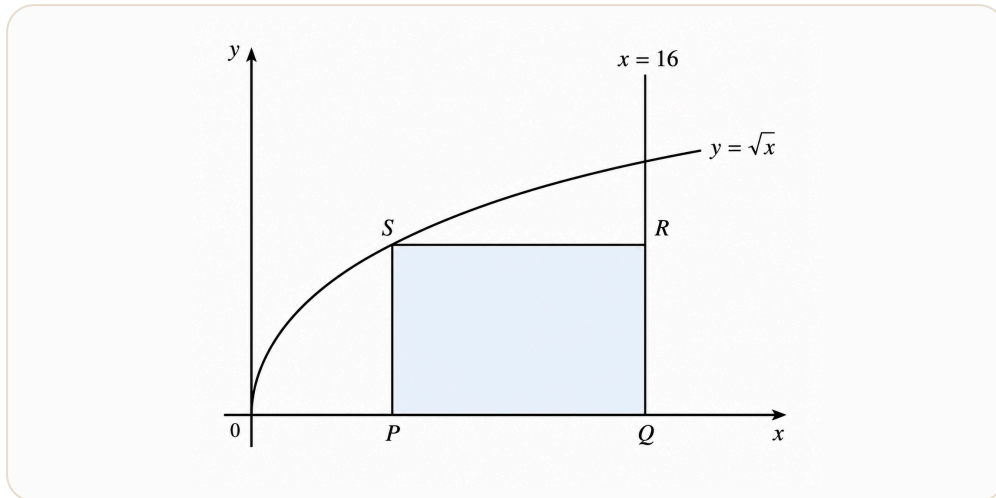
This simplifies to $(8r/5)/(1+r) = 3/5$.

So $8r = 3 + 3r$, hence $5r = 3$ and $r = 3/5$.

Then $a = (8/5)(1-3/5) = (8/5)(2/5) = 16/25$.

Therefore $a + r = 16/25 + 15/25 = \mathbf{31/25}$.

Question 25



PQRS is a rectangle. P and Q lie on the x-axis, Q and R lie on the line $x = 16$, and S lies on the curve $y = \sqrt{x}$, with $0 < x < 16$. What is the maximum possible area of the rectangle?

A $64\sqrt{3/9}$ B $128\sqrt{3/9}$ C $64\sqrt{3/3}$ D $128\sqrt{3/3}$ E $32\sqrt{3}$

WORKED SOLUTION

Let $S = (x, \sqrt{x})$. Then $P = (x, 0)$, $Q = (16, 0)$, and $R = (16, \sqrt{x})$.

The width of the rectangle is $16 - x$ and its height is $\sqrt{x}$, so

$A = (16 - x)\sqrt{x}$.

Let $t = \sqrt{x}$, so $x = t^2$. Then $A = (16 - t^2)t = 16t - t^3$.

Differentiate: $dA/dt = 16 - 3t^2$.

Set to zero: $16 - 3t^2 = 0 \Rightarrow t = 4/\sqrt{3}$.

Then $A_{\max} = 16 \cdot (4/\sqrt{3}) - (4/\sqrt{3})^3 = 64/\sqrt{3} - 64/(3\sqrt{3}) = 128/(3\sqrt{3}) = 128\sqrt{3}/9$.

Question 26

Find the maximum value of the gradient of the curve $y = 1 - 2x + 4x^{3/2} - x^2$, where $x > 0$.

A $\frac{1}{2}$

B 1

C $\frac{3}{2}$

D 2

E $\frac{5}{2}$

WORKED SOLUTION

The gradient is $dy/dx = -2 + 6\sqrt{x} - 2x$.

Let $t = \sqrt{x}$, where $t > 0$. Then the gradient becomes

$$g(t) = -2 + 6t - 2t^2 = -2(t^2 - 3t + 1).$$

This is a downward-opening quadratic, so its maximum occurs at $t = -b/(2a) = 3/2$.

$$\text{Then } g_{\max} = -2 + 6 \cdot (3/2) - 2 \cdot (9/4) = -2 + 9 - 9/2 = \mathbf{5/2}.$$

Question 27

The internal angles of a triangle are α , β and θ , and

$$3 \tan \alpha - 2 \sin \beta = 2,$$

$$5 \tan \alpha + 6 \sin \beta = 8.$$

What is the value of θ in degrees?

A 15

B 45

C 75

D 105

E 135

WORKED SOLUTION

Let $u = \tan \alpha$ and $v = \sin \beta$. Then we have

$$3u - 2v = 2 \text{ and } 5u + 6v = 8.$$

Multiply the first by 3: $9u - 6v = 6$.

Add to the second: $14u = 14$, so $u = 1$.

Then $3(1) - 2v = 2$, so $v = 1/2$.

Hence $\tan \alpha = 1$, giving $\alpha = 45^\circ$, and $\sin \beta = 1/2$, giving $\beta = 30^\circ$ or 150° . The value 150° is impossible because $45^\circ + 150^\circ$ exceeds 180° , so $\beta = 30^\circ$.

Therefore $\theta = 180^\circ - 45^\circ - 30^\circ = \mathbf{105^\circ}$.